

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

Currently, the clinic uses spreadsheets and physical paper records to manage details such as patients, practitioners, appointment bookings, date/time; making the process of booking appointments less efficient and prone to errors such as double booked appointments/redundancy, disorganized information etc. The client (clinic) needs a software to help manage these stakeholders like patients, practitioners and appointments, but does not specify or mention details such as appointment clashes, appointment/booking availability, whether or not patients enter/register and book appointments themselves or is it necessary for admin to book manually, and whether or not there are other branches of the clinic when booking. Solving or addressing these issues would help streamline the booking appointments process, support patient convenience, support overall clinic growth, and ease admin work. While design criteria is vague and not specific enough, one must assume all aspects to ensure system is not limited to vague requirements that offer no real function unless expanded on.

2. Initial stakeholders

Stakeholder	Possible need
Admin staff	Easy navigational system, streamlining schedule and booking management as well as patient details
IT support staff	Maintaining systems
Practitioners	Access to patient info, patient type, appointment type by need
Patients	Easy booking, retention reminders, accurate personal details (can change when needed)

3. Initial features

Feature	Confirmed or provisional?	Why?
Patient details	Confirmed	To know patients needs, specifically mentioned in terms of managing data
Schedule availability check	Confirmed	So there are no clashes
Booking the appointment (admin or patient but not both to reduce redundancy)	Confirmed	Main function of system
SMS reminders	Confirmed/provisional	Useful in making sure there are

		zero no-show appointments
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4. Questions for the client

1. Are there any current clashes with booking times like duplicate patient records or appointments?
2. Are appointments booked by admin staff over the phone with patients or its flexible?
3. What are your current security flaws/concerns
4. Can you be more specific on the design criteria?
5. Any other features you expect in the final or during the development of the system?

5. What we do not yet know

1. How the booking process works
2. Patient detail privacy and the clinic's legal commitments
3. Login portals for clinic staff vs landing page/basic booking page for patients.

AI Activity Card - Ask, Check, Explain

Before AI

A basic appointment booking system where you manually enter values as rows of code in the system/application. Covering bare details like time/date, patient name, and practitioner; nothing more.

Problems:

1. Both codes don't require user input (doesn't distinguish between SmartCare staff or patient input if patients made appointments themselves)
2. No room for checking invalid inputs
3. No appointment edit option
4. Appointment classification not there
5. ValueError specifies that patient name cannot be empty, but practitioner and appointment date/time can be? Also this points out that although patient name cannot be empty, there is no way of deciding the output of that as there is no way to enter any type of input.

AI request (**Used Microsoft Copilot as advised/instructed)

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

The program prints a welcome message, then displays two hard-coded appointments using simple variables. In the enhanced section, it introduces a list (appointments) and two functions:

- `book_appointment()` — creates a dictionary containing patient, practitioner and time, then adds it to the list. It also checks that the patient name is not empty.
- `display_appointments()` — prints all stored appointments, or a message if none exist.

Finally, it books two appointments and displays them.

- **No conflict checking** — the system allows two appointments at the same time with the same practitioner.
- **Weak validation** — only the patient name is checked; practitioner and time could be blank or None.
- **No user input** — everything is hard-coded, so the receptionist cannot actually enter appointments.

1. Why is a **list of dictionaries** more flexible than using separate variables like `patient1_name`, `patient2_name`, etc.?
2. What happens in your program if you call `book_appointment(None, 'Dr. John Doe', '10:00 AM')`?

**Used Microsoft Copilot as advised/instructed

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Identify and react to inputs	Yes			

accordingly				
Clarify design brief		Yes		
Program must handle multi-user requirements	Yes			
Streamline booking process		Yes		

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

Identify and react to inputs accordingly	Accept
Clarify design brief	Modify
Program must handle multi-user requirements	Unverified
Streamline booking process	Accept

Verify

- Run the code - Works
- Test normal input - Works
- Test unusual input – Does not work
- Compare with requirements - Works
- Ask tutor/peer
- Check documentation

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?

Yes I understand the code without reading the AI response, what I still need to understand is if or how I can make the code more simple in the sense that it matches requirements but also does more than that. For example, it addresses some of the issues I highlighted above, and it can have more assumed logic involved outside of what the given design brief is.